Numerical Study of Generic Convection-Diffusion Transport

Comparison of scalar transport in forced and recirculating flow fields

1. Aim

To examine how convection and diffusion influence the transport of a scalar temperature field in three prescribed velocity fields: a forced flat-plate flow, a forced channel flow and a natural-convection surrogate with recirculation. The comparison focuses on temperature distributions, thermal boundary-layer development and outlet or mean temperature at different Peclet numbers.

2. Computational Model and Cases

The scalar field is calculated on a two-dimensional Cartesian domain. Each case uses a different prescribed velocity field and corresponding thermal boundary conditions. The forced flat-plate case represents heating along a wall, the channel case represents heating from channel walls, and the recirculating surrogate uses hot and cold vertical walls to drive circulation.

Parameter	Description / value
Coordinate system	Cartesian
Grid points	65 × 65
Peclet numbers	50, 200 and 1000
Velocity fields	Forced flat plate; forced channel; recirculating surrogate
Scalar transport	Transient convection-diffusion equation
Convection discretization	Upwind
Diffusion discretization	Central difference

3. Governing Equation

The scalar transport equation used in the source report is:

$$\partial\phi/\partial t + \nabla\cdot(u\phi) = \nabla\cdot(\Gamma\nabla\phi) + S\phi$$

Here, ϕ is the transported scalar, u is the prescribed velocity field, Γ is the diffusion coefficient and $S\phi$ represents a possible source term. The cell Peclet number is defined as $Pe = |u|\,\Delta x / \Gamma$. It expresses the relative importance of convective transport compared with diffusion.

4. Boundary Conditions and Numerical Method

Thermal boundary conditions depend on the selected flow case. The flat-plate case uses a heated wall, the channel case uses heated channel walls, and the recirculating surrogate uses hot and cold vertical walls. The scalar equation is advanced using a finite-volume-style transient iteration, with upwind treatment for convection and central differencing for diffusion.

5. Quantities Examined

The source identifies residual behaviour, outlet or mean temperature, average wall heat flux, scalar boundary-layer thickness, local Peclet number and temperature-field shape as the quantities of interest. The available plots and tables provide approximate temperature profiles, case-averaged temperatures and flat-plate boundary-layer thicknesses.

6. Results at Different Peclet Numbers

The following table reorganizes the approximate values reported in the source. The boundary-layer thickness is specified only for the forced flat-plate case.

Measured quantity	Pe = 50	Pe = 200	Pe = 1000
Flat-plate outlet / mean temperature	≈ 0.24	≈ 0.097	≈ 0.032
Channel outlet / mean temperature	≈ 0.565	≈ 0.237	≈ 0.083
Recirculating-case mean temperature	≈ 0.50	≈ 0.50	≈ 0.50
Flat-plate layer thickness at x = 5	≈ 0.8–1.0	≈ 0.45	≈ 0.2
Channel centreline minimum temperature	≈ 0.18	≈ 0	≈ 0

7. Effect of Peclet Number on the Temperature Field

For the forced flat-plate case, the heated thermal layer begins at approximately x = 1.25 and grows in the downstream direction. Its thickness at x = 5 decreases from about 0.8–1.0 at Pe = 50 to 0.45 at Pe = 200 and 0.2 at Pe = 1000. The field plots on pages 3, 5 and 6 show the layer becoming progressively thinner and more closely attached to the heated wall as Peclet number increases. This is consistent with convection becoming more dominant relative to diffusion.

In the channel, the two heated-wall layers overlap more strongly at Pe = 50. The centreline minimum temperature is approximately 0.18. At Pe = 200 and 1000, the centreline temperature approaches zero, indicating that the two thermal layers have become separated by a colder core. The profile plots show sharper temperature gradients near the walls at the larger Peclet numbers.

The recirculating natural-convection surrogate behaves differently from the forced cases. Its mean temperature remains close to 0.50 for all three Peclet numbers. The profile plots show a relatively mixed core, with stronger gradients concentrated near the vertical walls, particularly at the higher Peclet numbers.

8. Comparison of Mean / Outlet Temperatures

The flat-plate value decreases from approximately 0.24 to 0.032 as Peclet number rises from 50 to 1000. The channel value also decreases, from about 0.565 to 0.083, but remains roughly 2.4–2.6 times the flat-plate value at each Peclet number. The source attributes this difference to the channel's two heated walls confining the warmed fluid, whereas the flat-plate arrangement is open on one side. The recirculating case stays near 0.50 because the hot and cold walls and circulation produce a nearly balanced mean temperature.

9. Conclusion

The simulations demonstrate that the three prescribed velocity fields produce distinct scalar-transport behaviour. In the forced cases, increasing Peclet number makes the thermal boundary layer thinner and reduces the outlet or mean temperature. The channel retains a higher mean temperature than the open flat-plate case, while the recirculating surrogate maintains a mean temperature close to 0.50 across the tested Peclet numbers.

The available results support comparison of temperature profiles, case-averaged temperatures and flat-plate layer thickness. However, wall heat flux, residual histories and local Peclet-number distributions are not included in the supplied figures, so quantitative comparisons of heat-transfer rate and convergence cannot be completed from this report alone.

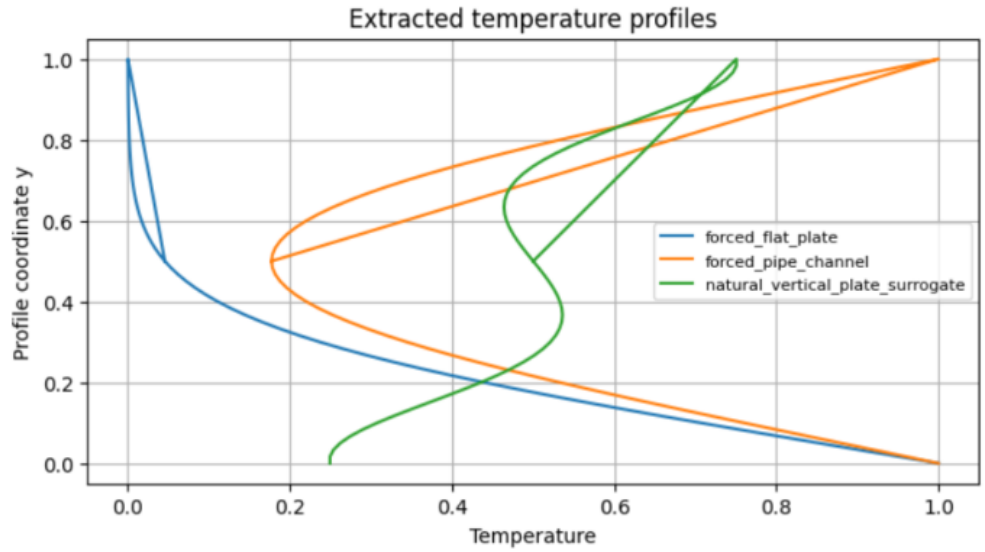
10. Data and Model Limitations

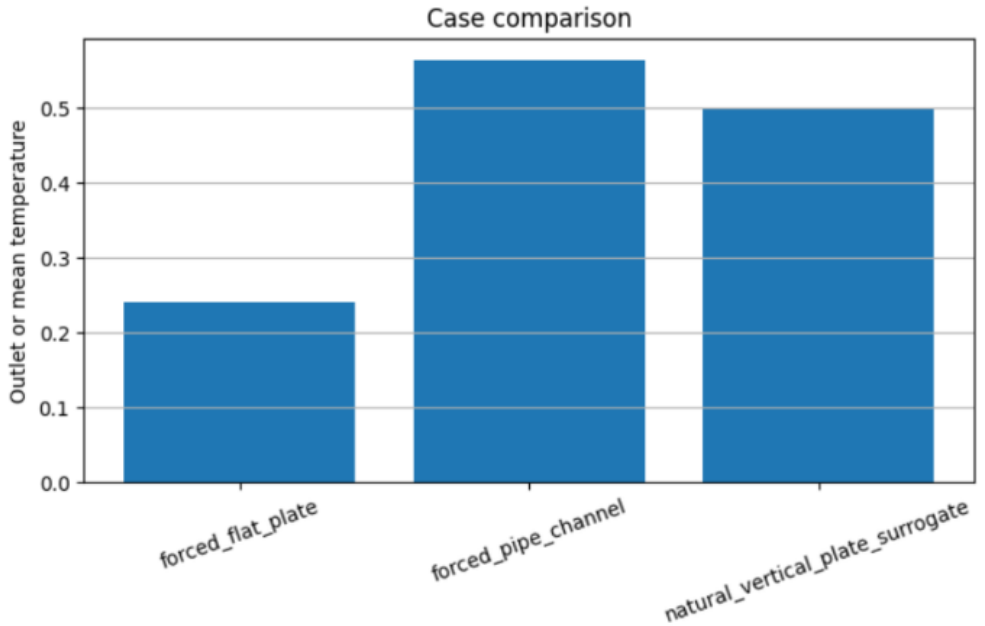
The original parameter table lists Reynolds number, lid velocity, CFL, SOR and a tolerance value that appear to belong to a separate lid-driven-flow setup rather than the scalar-transport cases. These entries have therefore not been used as governing parameters for the present comparison. The reported temperatures and boundary-layer thicknesses are approximate readings from the plots. Some extracted profile curves contain straight connecting segments between sampled portions; these have not been treated as physical temperature distributions.

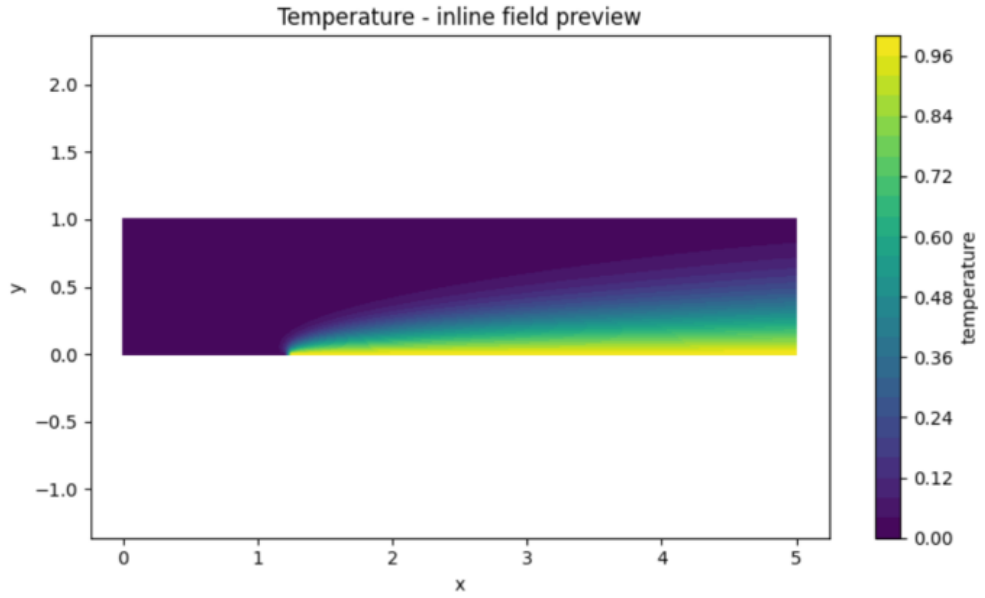
11. Original Simulation Figures

The original extracted temperature profiles, case-comparison charts and temperature contours are retained in the following pages.

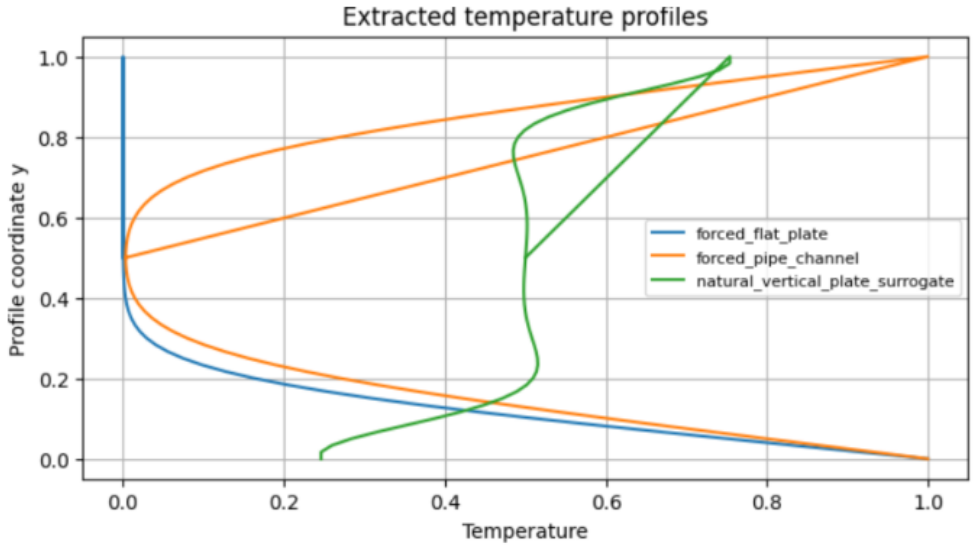
For Peclet number 50:

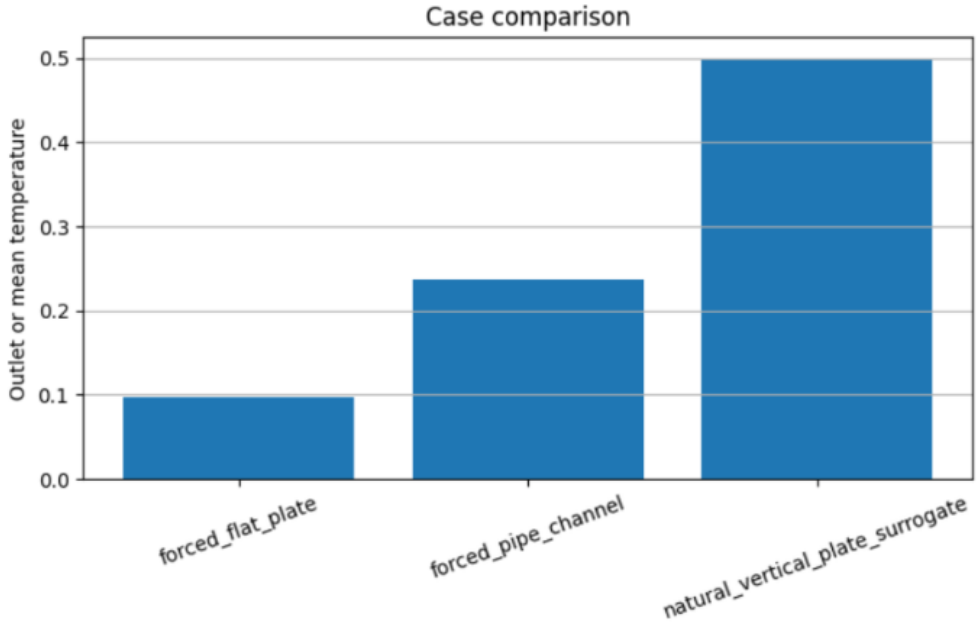


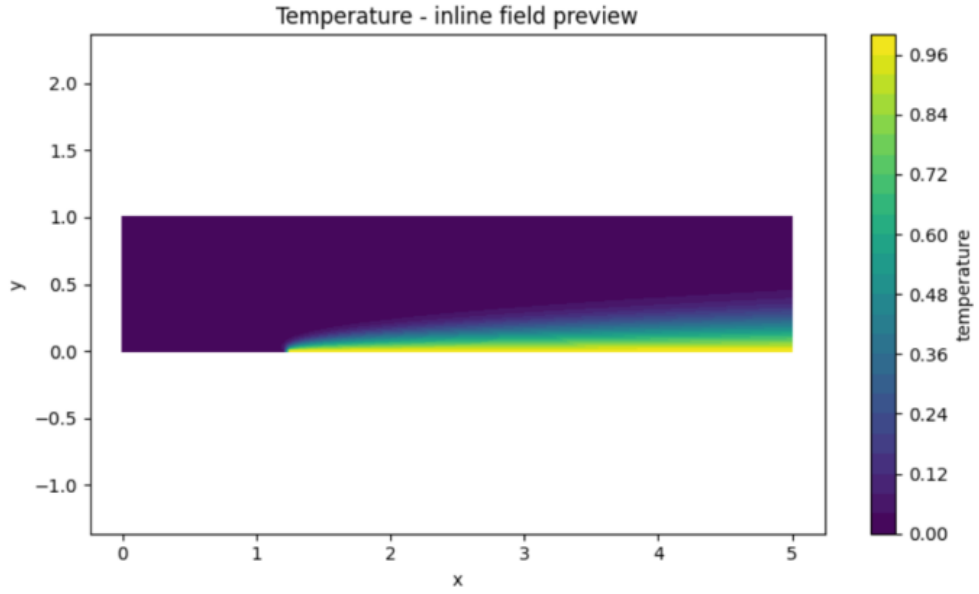




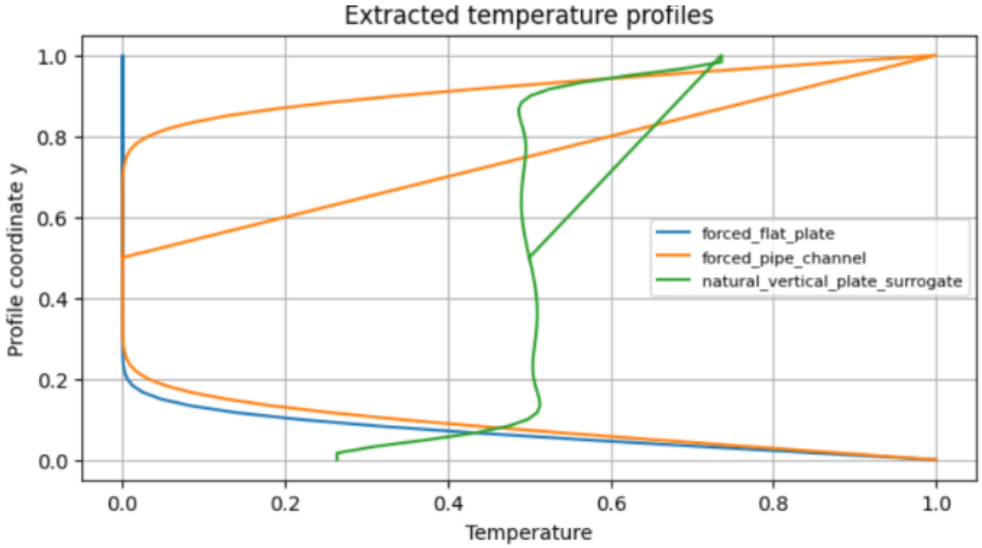
For Peclet number 200:

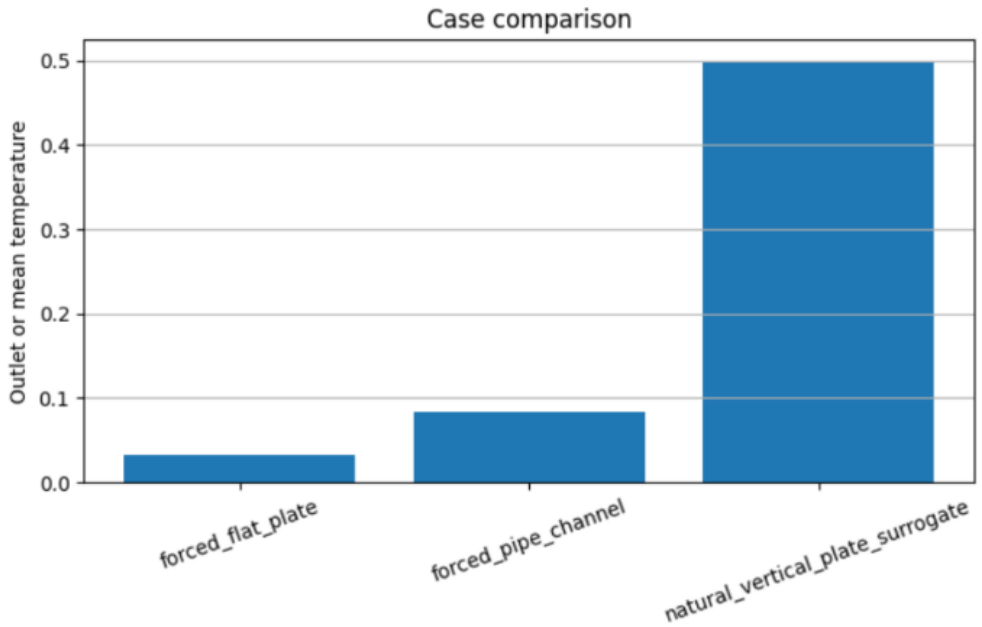


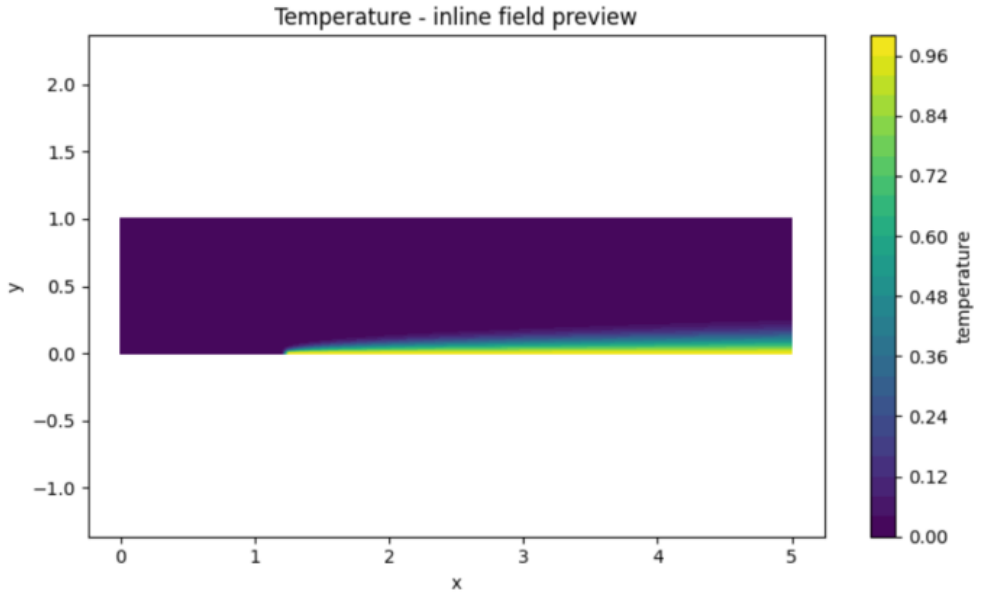

Case comparison
Outlet or mean temperature
0.0
0.1
0.2
0.3
0.4
0.5
forced_flat_plate
forced_pipe_channel
natural_vertical_plate_surrogate


Temperature - inline field preview
y
x
temperature
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
1
2
3
4
5
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

For Peclet number 1000:







Results:

Parameter	**Pe = 50**	**Pe = 200**	**Pe = 1000**
Outlet / mean temperature: forced flat plate	≈ 0.24	≈ 0.097	≈ 0.032
Outlet / mean temperature: forced channel	≈ 0.565	≈ 0.237	≈ 0.083
Outlet / mean temperature: natural-convection surrogate	≈ 0.50	≈ 0.50	≈ 0.50
Scalar boundary-layer thickness at x = 5 (flat plate)	≈ 0.8–1.0	≈ 0.45	≈ 0.2
Channel centreline temperature (profile minimum)	≈ 0.18	≈ 0	≈ 0

Parameter	Pe = 50	Pe = 200	Pe = 1000
Field shape, forced cases	Thick, diffusive layer	Thinner layer	Thin layer hugging wall

Conclusion:

Scalar transport by upwind convection and central diffusion was compared for three prescribed velocity fields: a forced flat plate, a forced channel and a natural-convection surrogate with hot and cold vertical walls. The cases were run at Peclet numbers of 50, 200 and 1000 on a 65 × 65 grid. In every case the temperature field is bounded between the wall values, and the forced cases show the expected thermal boundary layer growing downstream of the start of the heated wall.

Effect of Peclet number on field shape and boundary-layer thickness: For the forced flat plate, the heated layer starts at about x = 1.25 and grows downstream. At Pe = 50 it is thick and diffuse, reaching about y = 0.8–1.0 at x = 5. At Pe = 200 it is about 0.45 thick, and at Pe = 1000 it is a thin layer of about 0.2 attached to the wall. The thickness falls by roughly a factor of 2 for each four- to five-fold increase in Pe, which is consistent with the usual scaling $\delta \propto Pe^{(-1/2)}$. As convection dominates over diffusion, heat is carried downstream before it can spread across the stream, so the temperature gradient at the wall becomes steeper.

Outlet or mean temperature: The outlet or mean temperature of the forced cases decreases strongly with Pe. For the flat plate it falls from about 0.24 to 0.097 to 0.032, and for the channel from about 0.565 to 0.237 to 0.083, at Pe = 50, 200 and 1000. The channel is warmer than the flat plate by a factor of about 2.4–2.6 at every Pe, because its two heated walls are close together and confine the heated fluid, while the flat plate is open on one side. The natural-convection surrogate stays at about 0.50 for all three Pe, since its hot and cold walls are symmetric and the recirculating flow redistributes heat between them, so its mean temperature does not depend on Pe.

Temperature profiles: In the channel, the centreline temperature falls from about 0.18 at Pe = 50 to nearly zero at Pe = 200 and 1000, and the two wall layers separate, leaving a cold core between them. At Pe = 50 the layers from the two walls are still merging. The natural-convection profile is different from the forced ones. It has a nearly uniform core near 0.5, with thin layers near the walls at Pe = 200 and 1000, and a more strongly curved profile at Pe = 50. This is the behaviour expected when a recirculating flow mixes the core and leaves the gradient in the wall layers.

Limitations of the figures: The wall flux, the final residual, the convergence history and the local Peclet number were requested in the observations but are not shown in the figures, so wall-flux and convergence comparisons cannot be made from the results here. The boundary-layer thicknesses and the mean temperatures above are read from the field plots and bar charts, so they are approximate. The extracted-profile plots also contain straight connecting segments (for example near y = 0.5), which look like joins between separately extracted profiles and not physical temperature distributions, so they were not used for quantitative values. In addition, the parameter

table lists Reynolds number, lid velocity, CFL, SOR and a tolerance of 1e-0.8, which appear to belong to a different (lid-driven) case and not to this scalar-transport study.

Overall, the three velocity fields give clearly different scalar transport. In the forced cases the thermal layer becomes thinner and the outlet or mean temperature falls as Pe increases, with the channel retaining more heat than the open flat plate. The natural-convection surrogate gives a mean temperature of about 0.5 at all Pe and has a mixed core, so it is controlled by the recirculation and not by the Peclet number. Reporting the wall flux, residual history and local Peclet number would complete the comparison of heat-transfer rate and convergence.